

When Modal Contact Rows Fail in Fixed-Budget XPBD: Diagnosis and an Operating Guide

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Abstract

Rigid-body XPBD engines already resolve contacts and joints, and adding a small global modal state (16–24 amplitudes) is an attractive way to give a stiff prop low-cost vibration without converting it into a full nodal/tet deformable. We show that inserting this globally shared modal state directly into a two-way unilateral contact row is not automatically energy-safe when the host runs only a few local iterations per substep. Across three scenes, two relaxations and four schedules, *our tested XPBD implementation* overdraws the measured gross rigid-side supply by up to 4.4×10^7 J (a $2.2\text{--}4.5 \times 10^7$ J neighborhood over a deterministic perturbation ensemble), while augmented-Lagrangian and implicit implementations are markedly milder controls (≤ 6.7 J in 3 of 24 cells and never, respectively; a gravity/no-contact accounting audit places the control floor below 10^{-3} J, so 6.7 J is real injection). A same-path iteration ladder, equal-row-count schedules, warm-starting and mode truncation localize this catastrophic amplification to finite-budget convergence of the shared modal coupling, and an in-host block-condensation ablation shows condensing the shared rows does *not* fix it. When the contact architecture is flexible the tested stiffness-aware implicit realization is the preferred observed control; in a fixed XPBD host, spend the local budget on iterations and verify the modal basis — more substeps and a block solve do not suffice. For hosts that still cannot converge, a cumulative modal-storage projection enforces its scalar accounting bound unconditionally (Prop. 4.1, roundoff floor in all 90 measured cells) but opens up to 21.6 mm of penetration (72% of the board's thickness), and is therefore containment, not corrected contact.

CCS Concepts

• Computing methodologies → Physical simulation.

Keywords

contact, modal reduction, position-based dynamics, energy safety, fixed-budget solvers

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1 Introduction

A developer building an interactive scene on a rigid-body XPBD engine [11] already has a solver that resolves contacts and joints under a small, fixed local-iteration budget. To make a stiff prop — a shelf, a table, a ledge — ring when it is struck, the tempting move is not to replace it with a full nodal or tetrahedral deformable, which adds thousands of online coordinates and local internal constraints,

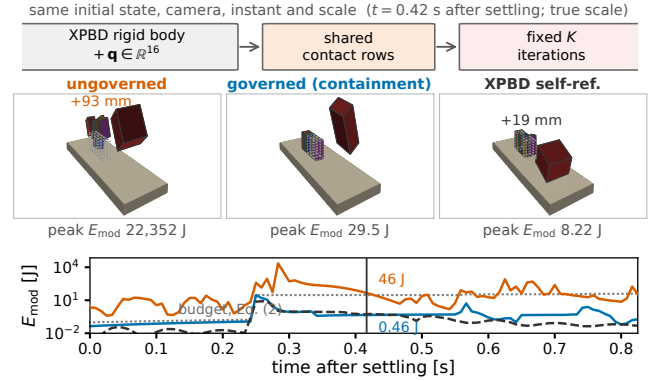


Figure 1: One shelf impact at a production-like interactive budget (1×8), true scale, three independent runs. Ungoverned, truncation energy throws the resting books up to 93 mm — more than their own height; governed, (2) holds peak modal energy to 29.5 J and they stay within 1 mm of rest; the reference is the host's own high-iteration self-reference (500×1). The bound removes the spurious launch and some legitimate motion with it: the reference lifts one book 19 mm, the governed run none.

but to attach a small precomputed modal basis: 16–24 global amplitudes $\mathbf{q}$ capturing the low-frequency bending of a stiff, small-displacement body [8, 9, 13]. The runtime state grows by a handful of numbers, the precomputation is offline, and the same XPBD host steps everything. This paper is written for that developer, not to argue XPBD against another solver.

The catch is in the contact. A modal prop driven *one way* — an external filter reads detected contact events and rings the modes with no back-reaction — is the real-time precedent [3], but it double-counts the interface energy and cannot let the prop's own deformation change how it is supported. The physically consistent move is a *two-way* row: rigid and modal unknowns resolved together through one shared contact multiplier [17], so a heavier load bends the support and the bent support redistributes the load. The trap is that every support contact touches the *same* global vector $\mathbf{q}$, so the contact-row graph is dense on the modes even though the state is tiny — and a host running only a few local iterations per substep need not converge that shared coupling. We take the row as given: at position level it is one linearization step from the velocity-level law ($\mathbf{C}^{n+1} \approx \mathbf{C}^n + h \mathbf{J} \mathbf{u}^{n+1}$, same Jacobian and multiplier as Zheng and James [17]), narrower than the original — normal rows only, $e=0$ restitution — and *not* a contribution of this paper. Our question is the practitioner's: what does the obvious two-way transcription cost at a normal budget, and what should one do about it? *If solver*

architecture is unconstrained our evidence favors the tested stiffness-aware velocity-level realization; our target is the existing fixed-budget XPBD host.

Finite-iteration coupling across a partition boundary can inject spurious energy even when every subsystem is individually passive [15], and velocity-level frictional contact spanning rigid and reduced deformable bodies, with the energy artifacts of iterative solvers named, goes back at least to Kaufman et al. [10]. What is missing, and what we supply, is a quantitative account of this case — unilateral contact-to-modal transfer in a fixed-budget host — and an operating guide for the practitioner who hits it.

Contributions.

- (1) **A practitioner workflow and its failure.** We treat adding a compact modal state to a fixed-budget XPBD host as the workflow it is, and show that the direct two-way contact row overdraws its measured rigid-side supply catastrophically at interactive budgets — across three scenes, two relaxations and four schedules (§3.1) — with augmented-Lagrangian and implicit implementations as milder controls. The cross-control reading is secured by a gravity/no-contact accounting audit and the failure’s robustness by a deterministic perturbation ensemble.
- (2) **Mechanism and an operating guide.** A same-path iteration ladder, equal-row-count schedules, warm-starting, mode truncation and an in-host block-condensation ablation localize the amplification to finite-budget convergence of the shared modal coupling (§3.2), and yield an ordered decision rule: the tested implicit realization when the architecture is flexible, otherwise iterations and a verified basis — not more substeps, not a block solve — in a fixed XPBD host.
- (3) **A last-resort guardrail, priced.** A cumulative modal-storage bound funded by the measured rigid-side loss and enforced by a scalar projection of the realized state (§4): a deliberately minimal fail-safe, not a contact method, guaranteed unconditionally (Prop. 4.1, roundoff floor in all 90 measured cells) but at a measured cost to contact validity (§4.1).

Related work. Rigid XPBD [11], interactive modal deformation and velocity-level modal contact [8–10, 17], and reduced XPBD [13] set the stage; the last uses modes as a *deformation* surrogate, whereas here modal amplitudes enter as contact DOFs driven by the shared multiplier (cf. ray-traced penalty contact without a shared unilateral multiplier [18]). Our last-resort guardrail transplants energy-tank control [5, 7, 14] rather than inventing one: we keep the tank-and-observer structure but fund the tank from *measured* rigid-side loss — not the port it regulates — and scale realized *state* rather than a force, because a fixed-budget solver commits its multipliers before the excess is observable. This separates it from total-energy projection [4], bidirectional total-energy tracking [16], and the structural finite-budget passivity of *bilateral* couplings [15].

2 The row and the invariant it must respect

Let $\mathbf{q} \in \mathbb{R}^r$ be mass-normalized modal amplitudes of the supporting body, so its modal mechanical energy is $E_{\text{mod}} = \frac{1}{2} \dot{\mathbf{q}}^\top \dot{\mathbf{q}} + \frac{1}{2} \mathbf{q}^\top \mathbf{K} \mathbf{q}$. A support-contact row between a rigid corner at world height y_c and the deformed surface has gap

$$C = y_c - (y_{\text{rest}} + \mathbf{U}_y^\top \mathbf{q}), \quad C \geq 0, \lambda \geq 0, C\lambda = 0, \quad (1)$$

with $\partial C / \partial \mathbf{q} = -\mathbf{U}_y$, so one multiplier λ drives both the rigid body and the modal amplitudes. Equation (1) is the transcription discussed above, not a new law. It is one row *law*, instantiated at every support contact every substep — 48 such rows on the shelf, 40 on the ledge, 200 on the table — all sharing the one modal state $\mathbf{q}$.

The invariant. We require that the energy stored in the modal subsystem never exceed the *measured gross rigid-side loss* — defined exactly at (3) below — cumulatively over the run:

$$E_{\text{mod}}^n - E_{\text{mod}}^0 \leq \eta \sum_{k \leq n} \max(\Delta E_{\text{rig}}^k, 0), \quad \eta \in (0, 1]. \quad (2)$$

The transfer efficiency η sets how much of the measured loss may be spent; *every experiment in this paper runs* $\eta = 1$, the least restrictive admissible value, so no result below depends on tuning it. The bound is *cumulative*, not per-step: a substep may deposit more than it earns provided the running total remains within the budget, which is what lets a legitimate ring persist across substeps.

Definitions. Modal storage is the mechanical energy of the reduced state as written above, and $E_{\text{mod}}^0 = 0$: every scene starts from an undeformed, resting support, so even the run’s own settling transient must be funded by the reservoir. The supply is

$$\Delta E_{\text{rig}}^k = (E_{\text{rig}}^{k,-} - E_{\text{rig}}^{k,+}) + W_g^k, \quad W_g^k = \sum_b m_b \mathbf{g}(\mathbf{x}_b^{k,+} - \mathbf{x}_b^{k,-}), \quad (3)$$

where $E_{\text{rig}} = \sum_b [\frac{1}{2} m_b \|\mathbf{v}_b\|^2 + \frac{1}{2} \boldsymbol{\omega}_b^\top \mathbf{I}_b \boldsymbol{\omega}_b]$ is *purely kinetic* — translational and rotational, over all dynamic bodies, not merely those in contact — and $\pm$ denote the two ends of substep k . We call this the **measured gross rigid-side loss**: the scene-wide, gravity-corrected, gross kinetic-energy loss of the rigid subsystem. It is a *loss*, not a dissipation — it also counts legitimate rigid-to-modal transfer and losses at rigid-rigid contacts elsewhere in the scene — so it is an envelope on what contact dissipated, not a measurement of it (§6). Gravity enters only through W_g , so free ballistic motion registers zero supply — a falling body’s kinetic gain exactly equals the gravity work, and only an inelastic velocity change funds the reservoir. Substep-boundary endpoints tile the timeline exactly, so no rigid energy change escapes the accounting.

What the invariant is, and two diagnostics. The bound (2) is a cumulative storage ceiling on the modal subsystem whose supply is *measured, not prescribed* — the right-hand side is the measured loss of (3), never a tuned constant or a fraction of incident energy. It constrains a scalar reservoir, not a contact port, so it is neither port passivity nor a signed per-interface bound. The *gross* sum ignores substeps in which the rigid subsystem *gains* energy, and counts energy returned and re-dissipated as fresh supply (§6 measures both). We read two diagnostics against the sweep below and keep them distinct: the incident-energy ratio R = peak modal energy / peak incident rigid KE, a severity measure, and the signed margin

of (2) itself in joules, the invariant proper; they need not flag the same cells. Enforcing the bound needs a projection, deferred to §4.

3 The failure at a fixed budget

All solver-behaviour measurements below come from a single machine (Apple M4, CPU, CPython 3.12); every number is frozen with its generating command and commit hash in the supplemental ledger. We deliberately avoid mixing machines: chaotic contact stacks diverge across architectures under floating-point reassociation.

3.1 One row, three implementations, the same 24 cells

We drive an impactor into a modally reduced support in three scenes — a shelf board, a ledge slab, and a 2.2×1.1 m table, the last two following the layouts of Coevoet et al. [3] — and sweep modal relaxation and the local budget (iterations $\times$ substeps) over $\{4 \times 1, 8 \times 2, 16 \times 4, 32 \times 8\}$. The hosts are XPBD [11], AVBD [6] (a descendant of vertex block descent [2]), and a sequential-impulse realization in the sense of Catto [1], made implicit in the modal block. All three are pinned to $h=1/120$ s, per-substep contact regeneration, modal rank 16/16/24 (shelf/ledge/table), Rayleigh damping, an *implicit-midpoint* modal restoring step, and $\eta=1$; they differ in the unknown solved for (position / position + multiplier / velocity), the modal contact weight — XPBD’s explicit per-mode compliance $1/(H_{ii}h^2 - 1)$ versus the impulse host’s implicit $(M+hD+h^2K)^{-1}$ — warm start, and relaxation (the full host/parameter table is in the supplement). Two differences matter. Relaxation is *inert* on the impulse host (its implicit modal weight needs no under-relaxation), so its two relaxation rows are bit-identical. Warm start also differs ($\lambda \leftarrow 0$ every substep for the position-based host, carried duals for the others), so the comparison is as-deployed, not compliance-matched — but it is not the lever: warm-starting the position-based host leaves the footprint unchanged (the same 8/24 cells, worst 1.20×10^5) and *worse* at 4×1 ($3.1 \times$), a warm multiplier being unable to converge the shared modal DOF that only iterations do (§3.2).

Equal row evaluations, unequal outcome. A cell costs $K \cdot S$ row evaluations per frame, so the budget axis spans $\times 64$ and conflates two refinements that are neither interchangeable nor equally priced (a substep also repeats contact generation and integration). At 32 row evaluations on the shelf drop, spent as iterations ($K=32, S=1$) the ratio is 0.300 and (2) holds; spent as substeps ($K=4, S=8$) it is 3.13, overdrawing by 481 J: substep refinement alone does not converge the row.

Figure 2 shows the outcome. The position-based host exceeds $R = 1$ in 8 of 24 cells, worst case 1.19534×10^5 (ledge, relaxation 1.0, 4×1 : peak modal energy 4.44×10^7 J against an incident 371 J); the augmented-Lagrangian host in 2 of 24 — both at the starved 4×1 corner of the table scene, worst 1.70 — so it is not unconditionally benign either. The ordering is scene-dependent and it inverts: worst-over-scenes at 4×1 the position-based host is five orders of magnitude more energetic in the ratio R (1.2×10^5 vs 1.70), yet on the table scene itself the augmented-Lagrangian host is the worse (1.18 and 1.70 against a position-based worst of 0.37). Neither host dominates across scenes.

The ratio is not the invariant. We also measure (2) directly on all three hosts, ungoverned, running the reservoir accounting live while forcing $\gamma \equiv 1$: every state write in the enforcement path is guarded by $\gamma < 1$, so the trajectory is bit-identical to an ungoverned run (the ratios above reproduce the independent sweep exactly). The two diagnostics do not flag the same cells — the position-based host violates (2) in 9 of 24 against $R > 1$ in 8, the augmented-Lagrangian host in 3 against 2, the implicit host in neither (0/24, worst margin at the roundoff floor) — so neither subsumes the other. What orders the failures is magnitude: the position-based overdraw reaches 4.4×10^7 J, seven orders past the largest augmented-Lagrangian one (6.7 J, at the starved 4×1 table corner) and past the implicit host’s exact zero. A gravity/no-contact accounting audit puts that control floor below 10^{-3} J on both non-injecting hosts — three to four orders under 6.7 J — so the augmented-Lagrangian overdraft is real impact-driven injection, not accounting noise: three regimes the ratio-to-incident-energy diagnostic compresses into one number.

3.2 The position-based amplification is a truncation artifact

Figure 3 isolates the local iteration budget. Two references recur below and are not the same object: *the implicit reference* is the implicit high-iteration realization at $K=500$ — the same code path run to $K=500$ — and *the XPBD self-reference* is the position-based host run to $K=500$ against itself. The position-based host falls monotonically from 2.96×10^4 at $K=1$ to 0.300 at $K=32$, approaching the implicit reference 0.2735; the gap $|\text{XPBD} - \text{reference}|$ shrinks from 2.96×10^4 to 2.7×10^{-2} . The implicit realization moves only in the fourth decimal across $K = 2 \dots 500$ (max–min 6.8×10^{-4}).

A falling energy ratio does not by itself show that the *row* is converging — a single scalar could shrink coincidentally. Measuring the constraint directly resolves the question, and the two complementarity conditions must be read in their own units rather than compounded into one dimensionally mixed scalar. End-of-substep penetration falls from 27.9 mm at $K=1$ to $3.6 \mu\text{m}$ by $K=64$, where it saturates. The multiplier surviving on *separated* rows — force where the gap is open, in units of the steady-state median — instead rises from $84 \times$ to $139 \times$ through $K=24$, collapsing to exactly zero only at $K=64$. The energy crossing at $K \approx 24$ therefore coincides with the *gap* converging, not with complementarity holding: the multiplier side clears a rung later. Position-based host only — the augmented-Lagrangian multiplier is not the same object, and the implicit realization is converged at $K=2$.

Self-convergence stops short of the reference. Running the position-based host to $K=500$ against itself does not close the gap: its ratio plateaus by $K \approx 64$ at 0.2996, so the residual 2.6×10^{-2} against the reference’s 0.2735 is a fixed point, not a tail. The state agrees no better — on the deflection trajectory $d_i(t) = U_y[i]^T q(t)$ the peak matches to 2.4% but the trajectories differ by 6.8 mm, 34% of the reference’s own peak. The two hosts converge to different discrete solutions, as expected when they discretize the same continuous law differently: the sweep thus separates a truncation pathology that convergence removes from a discretization difference that it does not.

This cuts two ways: convergence largely cures the amplification — iterating further or adopting the implicit realization removes it —

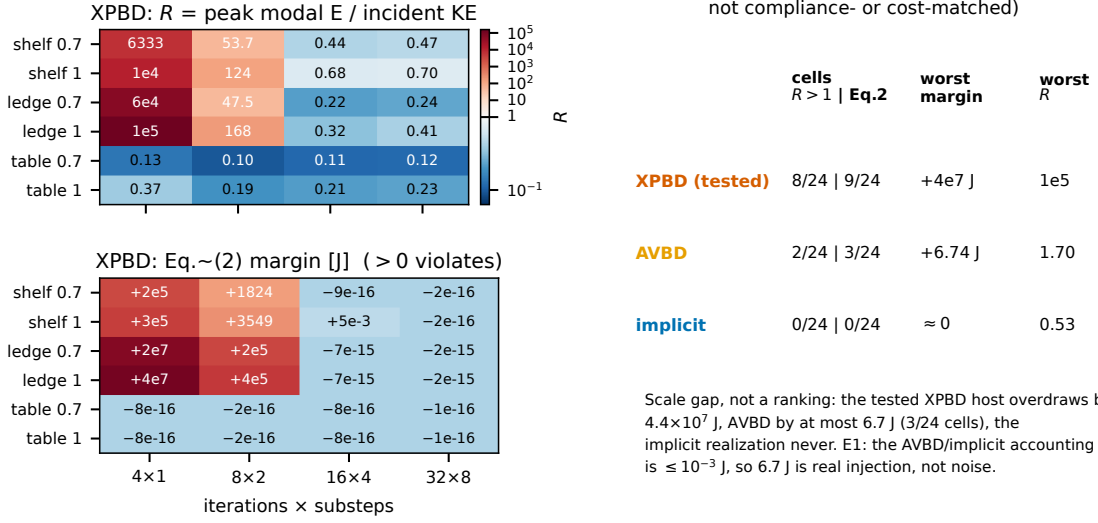


Figure 2: Our tested XPBD implementation, over the 24-cell sweep (3 scenes × relaxation {0.7, 1.0} × budget), governor OFF. Left, top: $R = \text{peak modal energy} / \text{peak incident rigid KE}$, a severity diagnostic; bottom: the signed margin of (2) in joules (> 0 violates). The two disagree by design (XPBD: $R > 1$ in 8/24, margin > 0 in 9/24) — a result, not noise. Right: a compact control strip for the three implementations (one of each, *not* compliance- or cost-matched): violating cells, worst margin, worst R . This is a scale gap, not a ranking — 4.4×10^7 J vs. ≤ 6.7 J vs. 0; the E1 accounting audit puts the control floor below 10^{-3} J, so 6.7 J is real injection. The full two-metric 72-cell three-implementation heatmap is in the supplement.

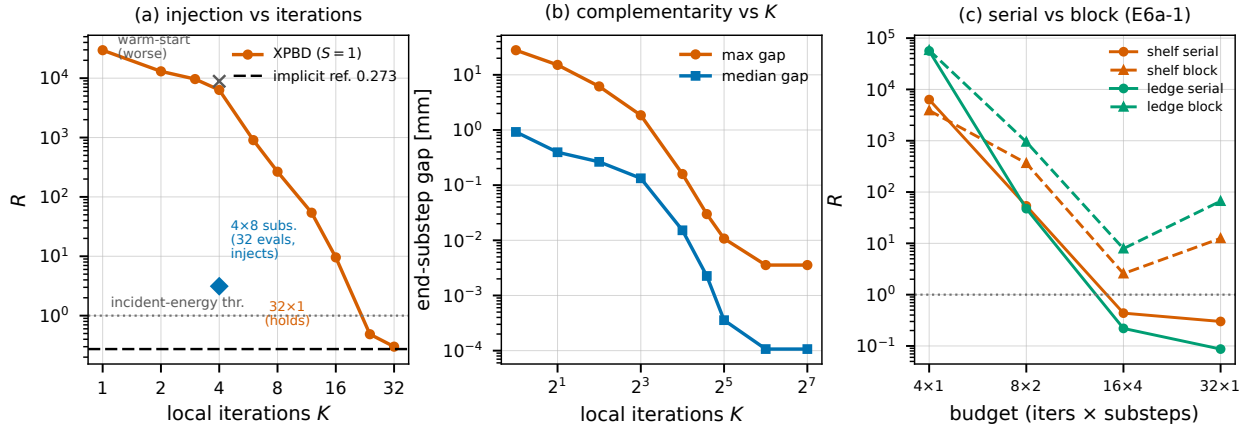


Figure 3: The XPBD operating envelope (our tested position-based host, governor OFF). (a) Injection R vs. local iterations K on a deterministic shelf drop ($S=1$): R decays monotonically toward the implicit reference (0.273), crossing the incident-energy threshold between $K=16$ and 24. Annotated: the 32×1 -vs- 4×8 equal-row pair (same 32 row evaluations, but spent as substeps it still injects) and warm-starting (no help; worse at 4×1). (b) The end-of-substep gap converges with K , confirming the falling ratio is the row converging, not a scalar shrinking coincidentally. (c) Shared-row treatment (E6a-1): serial per-row support projection (the paper path) vs. joint block condensation, both scenes. The serial path converges/holds by 16×4 – 32×1 ; block condensation does *not* — a different, worse convergence path that injects where serial holds. Condensing the shared rows is not the fix.

so the bound of §4 is not a substitute for convergence but targets the regime where neither escape is available. Interactive position-based solvers run schedules near 1×8 – 2×4 (the one-iteration small-step regime [12]), well below the $K \geq 24$ this scene needs: on 18 deployed

cells (3 scenes × 3 hosts, relaxation 0.7) the position-based host violates (2) in all six of its cells (worst $R = 2282$, on just 8 row evaluations) while the others match the sweep (1/6, 0/6).

Condensing the shared rows is not the fix. What converges the row is the iteration budget, not how the shared support rows are scheduled: replacing the serial per-row support projection with a joint block condensation of the active rows (modal weight, implicit-midpoint restoring step, budget and warm start held fixed; Fig. 3(c)) does *not* remove the injection. It is comparable at the starved 4×1 corner but markedly worse at every higher budget where the serial path holds ($R = 12.5$ and 66.5 at 32×1 on shelf and ledge, against serial's 0.30 and 0.09). Reproduced over two scenes and four budgets, it is a causal probe — a different, worse convergence path (diagonal rigid-body block approximation) — not a cure.

Not a knife-edge. The catastrophic cells survive a predefined local neighborhood. For each worst cell, the equal-row pair and the guardrail cell we run 16 paired deterministic perturbations — 12 physical (drop height $\pm 1\%$, normal speed, contact offset ± 1.5 mm, tilt) and 4 contact-row-order permutations, kept as separate cohorts — and every injecting cell overdraws in all 16, the sign never flipping, with a magnitude spread under one order of magnitude ($\log_{10}$ range $0.31\text{--}0.70$); the 4.4×10^7 J worst case is the maximum of a $2.2\text{--}4.5 \times 10^7$ J band. The equal-row inversion (32×1 holds, 4×8 injects, all 16) and the scene dependence (never injecting on the table) survive too. Mode truncation localizes but does not remove it: excluding the stiff cluster cuts the shelf 4×1 ratio $6333 \rightarrow 22.1$ (584 J still overdrawn) and 6 of 8 injecting cells still overdraw (worst ledge $+1.24 \times 10^6$ J), so band-limiting the co-solved basis is not sufficient.

4 A last-resort guardrail

When a fixed host cannot converge the row and neither the implicit realization nor more iterations are available, a deliberately minimal fail-safe holds (2) regardless of the contact solve: it credits a reservoir from the measured supply and, when the realized modal state would overdraw it, projects that state back inside.

Enforcement. A reservoir B accumulates the supply and is debited by realized storage. Both happen *after* the contact solve, within the same substep: ΔE_{rig} is not observable until the velocity solve has finalized the rigid velocities, so the substep credits $B \leftarrow B + \eta \max(\Delta E_{\text{rig}}, 0)$ at that point, tests the realized modal energy against it, and only then debits $B \leftarrow \max(B - \max(\Delta E_{\text{mod}}, 0), 0)$. If the realized modal energy would overdraw B , the state is projected onto the admissible set by a single scalar,

$$(\mathbf{q}, \dot{\mathbf{q}}) \leftarrow \gamma (\mathbf{q}, \dot{\mathbf{q}}), \quad \gamma = \min\left(1, \sqrt{\frac{E_{\text{mod}}^- + B}{E_{\text{mod}}^+}}\right), \quad (4)$$

with $E_{\text{mod}}^\pm$ the modal energy before and after the solve. Each substep runs identically in all three hosts: snapshot $E_{\text{rig}}^-, E_{\text{mod}}^-$; predict and run the contact solve; form ΔE_{rig} ((3), post-solve) and credit $\eta \max(\Delta E_{\text{rig}}, 0)$ to B in the *same* substep; if $E_{\text{mod}}^+ > E_{\text{mod}}^- + B$, scale the state by γ ; debit B — contact is **not** re-solved. The form (4) is closed because E_{mod} is quadratic: scaling $\mathbf{q}$ and $\dot{\mathbf{q}}$ together scales both terms by γ^2 , so it bounds total energy, not velocity alone, and is inert whenever the solve is within budget ($\gamma=1$, bit-identical trajectories).

PROPOSITION 4.1. *For any $\eta \in (0, 1]$ and any host, the loop maintains $B \geq 0$ and guarantees (2) unconditionally, assuming nothing*

Table 1: Post-projection contact validity for the *position-based* host at starved budgets, relaxation 0.7, per clamp-active substep. Gap violation is measured immediately after the γ -scale; the corrective impulse is the next substep's normal impulse against the unclamped steady-state median, a ratio because the hosts' multipliers are different objects. The 4×1 cells clamp on nearly every substep, so no steady state exists there (—).

scene	budget	clamped	gap viol. med / max [mm]	impulse	λ var.
shelf	4×1	107/108	1.24 / 21.6	—	—
shelf	8×2	171/216	0.21 / 6.43	3.68×	5.6×
ledge	4×1	97/108	0.08 / 21.2	1.03×	1.08×
ledge	8×2	178/216	0.07 / 4.08	8.92×	62×

about the contact solve, up to an accounting tolerance ϵ over n substeps (ϵ the per-test tolerance, 10^{-12} J).

Proof sketch. Carry two invariants over substeps, with $c_k = \eta \max(\Delta E_{\text{rig}}^k, 0)$: $B_k \geq 0$ and $(E_{\text{mod}}^k - E_{\text{mod}}^0) + B_k \leq \sum_{j \leq k} c_j$ (both hold at $k=0$). Crediting c_k *before* the test caps realized storage at $E_{\text{mod}}^- + B_{k-1} + c_k$ (as E_{mod} is quadratic), so $\Delta E_{\text{mod}} \leq B_{k-1} + c_k$; the debit keeps $B_k \geq 0$, and both invariants follow by induction. Since $B_n \geq 0$ this is (2) itself, not a relaxation — the same-substep credit absorbs the granularity term a naive telescoping would leave.

The governor holds. Turned on, all 72 sweep executions (60 distinct configurations, the implicit host's relaxation pairs bit-identical) and the 18 deployed cells of §3.2 satisfy (2) — 90 governed cells at the roundoff floor (worst margin 1.1×10^{-13} J). The worst realized ratio falls to 1.22 (position-based) and 0.87 (augmented-Lagrangian); the projection is inert wherever the solve is already within budget — every implicit cell, and 16×4 up on the position-based host at relaxation 0.7.

4.1 The cost of enforcement

Accuracy. Bounding the energy is useful only if the bounded solution improves on the unbounded one. At a moderate injecting cell (shelf, 8×2, relaxation 0.7) peak modal energy is 1555.6 J ungoverned and 29.56 J governed against the implicit reference's 7.92 J: the energy error falls $196\times \rightarrow 3.7\times$, and $189\times \rightarrow 3.6\times$ against the XPBD self-reference, so it does not depend on the reference. The trajectory, however, degrades — $\|d - d^{\text{ref}}\|_\infty$ goes from 6.5 mm (33% of the reference peak) to 14.2 mm (71%): the governor is a safety envelope, not an accuracy device. At the deployed 1×8 cell energy gives $2823\times \rightarrow 3.7\times$ and the trajectory improves ($108 \rightarrow 96\%$ of reference peak), though 96% is still not faithful.

This is why a 196× energy error coexists with a peak deflection only 1.2× too large: 99.6% of the excess sits in the shelf's stiff mode cluster above 20 kHz ($\sim 200\times$ the substep rate), which a single scalar cannot separate from the legitimate low-frequency sag — γ removes the *amount* but not the character ($99.6 \rightarrow 97.7\%$ above the cut), scaling the sag (24.1 $\rightarrow$ 5.9 mm against 20.0) down with the noise. The first-line remedy is band-limiting the basis, not a scalar.

Contact validity. Since $\gamma < 1$ shrinks the sag $\mathbf{U}_y^T \mathbf{q}$, (4) lifts the support *into* the resting body without re-solving contact. Table 1 quantifies the consequence, instrumented by wrappers that leave

the solve unperturbed (clamp counts reproduce the independent sweep exactly).

Median post-projection penetration is 0.07–1.24 mm, but the worst case reaches 21.6 mm, a $3.1\text{--}12.5\times$ increase over its pre-scale value — 72% of the 30 mm board thickness, and $1.05\text{--}1.08\times$ the board's peak dynamic deflection: a visible interpenetration where the projection is most load-bearing. Contact recovers on the next substep, but at a normal impulse up to $8.9\times$ the steady-state value and multiplier variance up to $62\times$ higher; rigid momentum is unchanged (exactly zero, verified), since (4) writes only modal state. On the augmented-Lagrangian host the projection is an order of magnitude gentler — at its one materially-overdrawing cell ($+6.7$ J) it opens 1.8 mm against 21.6, a single clamped substep, no measurable corrective impulse (full rows, and a decision table over these options, in the supplement).

5 Fidelity and runtime

5.1 The reduced response against a full-FEM reference

Modes are worth using only if the reduced response is faithful. Against an unreduced FEM solve on the *same* discrete operator (so discretization bias cancels), the reduced ledge response tracks the reference: the peak-deflection ratio rises $0.38 \rightarrow 0.54 \rightarrow 0.79 \rightarrow 0.88$ as h refines $1/120 \rightarrow 1/960$ (the residual $\sim 10\%$ is $k=24$ truncation, not coupling error), the ring frequency agrees to 0.4% (78.0 vs 78.3 Hz), and the far-field falloff tracks at Spearman $\rho = 0.89$ with no fitted attenuation.

5.2 Runtime cost

On the CPU host the ledger adds 0.23–0.41 ms at $16\times 4 - 0.9\text{--}3.4\%$ on an 11–126 ms baseline (the table scene's cost sits below its run-to-run variance, so we report the percentage). It is larger where the budget is tight: at the deployed $1\times 8/2\times 4$ schedules the same $\sim 0.3\text{--}2.2$ ms is 6.6–34.3%. We make no unqualified real-time claim; an *enforced* device-resident γ does not exist here (§6).

6 Limitations

Stability is not accuracy. Where the projection is load-bearing it opens contact violations up to 21.6 mm and moves the trajectory away from the implicit reference even as it moves the energy toward it (§4.1): bounded, not faithful. Its honest use is a safety envelope for budgets that would otherwise diverge; convergence is the better remedy wherever affordable. A mechanism scaling deviations about the quasi-static equilibrium rather than the whole state, preserving the load-bearing sag, is unbuilt.

Scope of the row. The coupling is normal-only — friction stays body-local, so scraping and rolling transfer lie outside it — and the position-level form fixes $e=0$. The bound governs modal storage only, not energy created spuriously in rigid DOFs: in the impulse realization, box–box rows rectify a support ring into net rigid motion, to which the ledger is blind by design.

Supply and coverage. The supply is a *gross* sum, so energy returned to the rigid bodies and re-dissipated is credited twice; we bound the exposure by the opposite channel $\sum_k \max(-\Delta E_{\text{rig}}^k, 0)$ as a

fraction of supply — 3–32% (position-based), 0.4–27% (implicit), 102–118% on the augmented-Lagrangian host (whose rigid subsystem gains more than it loses every cell, the box–box effect above) — all above the $\leq 1\%$ expected, and partition-dependent, so the budget belongs to the schedule as well as the scene. Enforcement is host-side and single-machine (CPU float64); a device-resident governor and a *governed*-path FEM validation remain future work. Three formulations, one implementation of each; three scenes, one contact regime — the augmented-Lagrangian host still overdraws in 3/24 cells (up to 6.7 J), so any claim that descent-class solvers are passive here is empirical about this sweep, not a theorem.

7 Conclusion

For a developer who has already chosen a fixed-budget XPBD rigid-body host and wants low-cost vibration on a stiff prop, the decision is ordered, not XPBD versus another solver. First pick the representation: full-space XPBD for large, local or nonlinear deformation, a compact modal state for stiff, small-displacement ringing. If the contact architecture is still open, the tested stiffness-aware implicit realization is the cleanest control we observed (it never overdraws), implementation-specific and not a solver-class theorem. If the XPBD host is fixed, do not tune blindly: the direct two-way modal row overdraws its measured supply by up to 4.4×10^7 J at a few local iterations, and neither more substeps at equal row cost, nor warm-starting, nor condensing the shared support rows into one block removes it (§3.2); converging the row with iterations does, so spend the budget there and verify the basis. Only when a fixed host still cannot converge does the cumulative modal-storage bound earn its place — it holds in all 90 measured cells but opens up to 21.6 mm of penetration, so it is containment, not corrected contact. A low-dimensional modal extension is cheap in state size; it is not automatically easy for a truncated local contact solve.

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